SCHOOL OF APPLIED SCIENCE AND OPEN LEARNING

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF CYBER SECURITY

FRAUD AND DETECTION TECHNOLOGIES

CYS 339

SESSIONAL EXAMINATION

DATE: 25-Oct-24 TIME: 14.00 - 17.00 HRS, 3 HOURS

Instructions to candidates

1. Write your **Name** and **Student ID** on the **Answer Sheet** provided
2. There are **Seven (7) questions** in this paper. Attempt any **five (5).** Indicate on the Answer sheet your choice from the optional questions.
3. Write your Answers on the answer Sheet provided.
4. No Foreign material will be allowed in this examination.
5. The number of marks is shown in brackets ( ) at the end of each question or part question.
6. Clear handwriting and neat work is encouraged.
7. The total number of marks for this paper is **100**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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QUESTION 1

a) After your graduation, you started working as a cyber security officer at Zamnet and your boss tasked to write a report on the measures the bank can undertake to prevent fraud. Write a report, in point form, outlining five (5) the bank can take. (10 marks)

b) Explain four (4) white collar crimes in relation to fraud. (4 marks)

c) Using two (2) examples in each case, differentiate between internal and external threats to a fraud detection system. (6 marks)

[Total: 20 marks]

QUESTION 2

The Law provide a formal mechanism for the creation and maintenance of social order in complex societies. They define crime, whether a crime has been committed, whether a given person is guilty of the act, and the convicted offender's punishment. The level of "evilness" is the basis of crime. We can distinguish crimes from one another in terms of the length of punishment and place of confinement. This is called grading. Crimes are generally graded into three categories:

a) Felonies, (8 marks)

b) Misdemeanours (6 marks)

c) Infractions (6 marks)

With the use two (2) of examples, discuss these types of crime

[Total: 20 marks]

QUESTION 3

a) With use of one (1) example, differentiate between Authorization and Accounting features of security. (6 marks)

b) According to research, criminals pursue their self-interests, maximising pleasure and reducing pain, such as material gain, personal safety, and social reputation, and make enemies without caring if they harm others in the process. Do you support or dispute this statement, state reasons for your opinion? (6 marks)

c) Explain any four (4) threats to a fraud detection system. (8 marks)

[Total: 20 marks]

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QUESTION 4

Cyber security analysts usually employ more than one of these techniques to help ensure ar accurate and comprehensive investigation. These are the specific methods analysts use for collecting data about requirements. These include:

i) Interview, (4 marks)

ii) Questionnaire, (4 marks)

iii) Record inspections (on-site), (4 marks)

iv) Research, (4 marks)

v) Sampling. (4 marks)

Discuss these methods, giving one (1) example in each case.

[Total: 20 marks]

QUESTION 5

a) Data Analysis is the process of systematically applying statistical and/or logical techniques to describe and illustrate, condense and recap, and evaluate data. There are two (2) common types of Data analysis. Using one (1) example, discuss these methods. (6 marks)

b) Describe the following functions used in data analysis

i) Pivot tables (2 marks)

ii) Vlookup (2 marks)

iii) Sorting functions (2 marks)

iv) Sum functions (2 marks)

v) If function (2 marks)

c) According to research, poverty among communities increases the likelihood of criminal behavior such as fraud in business organisations. Discuss this statement. (4 marks)

[Total: 20 marks]

QUESTION 6

a) As an Information Technology investigator, explain how you would go about to perform the following activities:

i) Capture live data (4 marks)

ii) Recover deleted data (4 marks)

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iii) Eradicate malware (2 marks)

b) Explain any two (2) reasons why investigators are advised not to turn off routers and other devices during an investigation process. (4 marks)

c) Discuss the process of securing evidence and reporting an incident to law enforcement agencies. (6 marks)

[Total: 20 marks]

QUESTION 7

a) Cyber Crime investigation process typically involves the following stages:

i) Identifying the cyber-crime (2 marks)

ii) Gathering the evidence (3 marks)

iii) Building a chain of custody (3 marks)

iv) Analyzing the evidence (2 marks)

v) Presenting the evidence (2 marks)

vi) Testifying (2 marks)

vii) Prosecution (2 marks)

Describe these stages

b) Discuss the process of gathering evidence on computers hidden partitions. (4 marks)

[Total: 20 marks]

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